

Introduction to Decentralized Finance - DeFi

Homework 5

1. Consider the MakeDAO protocol. An investor owns 20 ETH, each worth 1,000 Dollars. The investor deposits 6 ETH in a Vault in order to mint DAI (each worth 1 Dollar). The minimum collateralization ratio is 300%.
 - (a) What is the maximum number of DAI that the investor can mint?
 - (b) If the investor chooses to mint 1,500 DAI, what is the actual collateralization ratio?
 - (c) Having minted 1,500 DAI, what is the ETH price below which the loan becomes undercollateralized? Undercollateralized means that the collateralization ratio is smaller than the minimum.
 - (d) Suppose that the ETH price drops to 600 Dollars, how many ETH need to be added to the initial deposit of 6 ETH in order to avoid being undercollateralized?
 - (e) If the ETH price continues to be 600 Dollars, how many more ETH need to be added to the original deposit of 6 ETH in order to have an effective collateralization ratio of 400%?

2. An investor deposits 1000 DAI in Compound. DAI has a collateral factor of 90%.
 - (a) What is the collateralization ratio?
 - (b) If $1 \text{ DAI} = \$1$, what is the maximum dollar amount that the investor can borrow of any other asset in Compound?
 - (c) Suppose the investor also deposits 1 ETH valued at $1 \text{ ETH} = \$1,000$ and the collateral factor for ETH is 65. What is the total dollar value of the deposits (from both DAI and ETH deposits)?
 - (d) What is the collateralization ratio for the total deposits (from both DAI and ETH)?
 - (e) What is the maximum borrowing in dollars?

3. Suppose that the total supply in Compound is 200 million DAI. The total borrowed amount in Compound is 140 million DAI. The base rate is 0.25% ($a=0.0025$) and the slope is 10% ($b=0.1$).
 - (a) What is the utilization rate?
 - (b) What is the borrow rate?
 - (c) What is the supply rate?

4. The total supply in Compound is 200 million DAI. The total borrowed amount in Compound is 180 million DAI. The base rate is 0.5% ($a=0.005$). There is a kink in the slope at 0.8 (80%) utilization rate. The slope for utilization rates smaller than 80% is 10% ($b=0.1$). The slope for utilization rates bigger than 80% is 40% ($b=0.4$).
 - (a) What is the utilization rate?
 - (b) What is the borrow rate?
 - (c) What is the supply rate?

5. Suppose that the price of ETH is 1,000 DAI. An investor supplies 200 ETH in Compound and borrows 80,000 DAI. The borrowed amount in DAI is used to purchase an additional 80 ETH supplied to Compound. The borrow interest rate in DAI in Compound is 10%. However, the borrow interest rate in Aave is 6%.
 - (a) How can the investor use a flash loan in Aave to refinance the Compound loan with an Aave loan? List each of the steps.

6. In Uniswap there are 2,000 DAI (asset A) and 2,200 USDC (Asset B).
 - (a) What is the value of x ? What is the value of y ? What is the value of the invariant k ?
 - (b) An investor sells 50 DAI for USDC. How many USDC will the investor receive with the sale of 50 DAI?

7. Consider the DAI/USDC market in Uniswap with a supply of 110,000 DAI and 100,000 USDC. The exchange rate in Uniswap is 1.1 DAI for 1 USDC and the invariant is 11 billions. On a different Decentralized Exchange (DEX), the price for USDC is 1.05 DAI.
- (a) How would a trader arbitrage with a flash swap? Describe the individual steps assuming that the transaction involves either 1,000 DAI or 1,000 USDC.
 - (b) What is the profit earned by the trader? In answering this question assume that there are zero transaction fees.